Dataset

985, 978, 435, 389, 79, 926, 299, 538, 571, 828, 681,
302, 13, 518, 873, 256, 899, 864, 314, 470, 547, 440,
699, 867, 860, 202, 155, 792, 64, 406, 906, 859, 584,
375, 996, 466, 401, 428, 714, 453, 194, 487, 993, 34,
829, 317, 865, 296, 197, 895, 208, 613, 98, 487, 963, 81,
808, 182, 5, 869, 291, 549, 489, 49, 941, 473, 116, 705,
340, 209, 547, 156, 735, 573, 234, 259, 704, 711, 892,
509, 680, 280, 819, 385, 618, 666, 599, 389, 229, 862,
288, 971, 656, 18, 774, 226, 990, 786, 828, 605

any distribution



Understanding the central limit theorem

Sampling

985, 978, 435, 389, 79, 976, 299, 538, 571, 828, 681,
302, 13, 518, 873, 256, 892, 864, 314, 470, 547, 440
699, 867, 860, 202, 155, 792, 64, 406, 906, 859, 384,
375, 996, 466, 401, 428, 714, 453, 194, 487, 993, 34,
829, 317, 865, 296, 197, 895, 208, 613, 98, 487, 963, 81,
808, 182, 5, 869, 291, 549, 489, 49, 941, 473, 116, 705,
340, 209, 547, 156, 735, 573, 234, 259, 704, 711, 892,
509, 680, 280, 819, 385, 618, 666, 599, 389, 229, 862,
288, 971, 656, 18, 774, 226, 990, 786, 828, 605

Mean of sample k

$\bar{x}_1 = 555.2$

Sampling

985, 978, 435, 389, 79, 926, 299, 538, 571, 828, 681,
302, 13, 518, 873, 256, 899, 864, 314, 470, 547, 440,
699, 867, 860, 202, 155, 792, 64, 406, 906, 859, 584,
375, 996, 466, 401, 428, 714, 453, 194, 487, 993, 34,
829, 317, 865, 296, 197, 295, 208, 613, 98, 487, 963, 81,
808, 182, 5, 869, 291, 5[illegible], 489, 49, 941, 473, 116, 705,
340, 209, 547, 156, 735, 573, 234, 259, 704, 711, 892,
509, 680, 280, 819, 385, 618, 666, 599, 389, 229, 862,
288, 971, 656, 18, 774, 226, 990, 786, 828, 605

Mean of sample k

$\bar{x}_1 = 555.2$

$\bar{x}_2 = 439.5$

Sampling

985, 978, 435, 389, 79, 926, 299, 538, 571, 828, 681,
302, 13, 518, 873, 256, 899, 864, 314, 470, 547, 440,
699, 867, 860, 202, 155, 792, 64, 406, 906, 859, 584,
375, 996, 466, 401, 428, 714, 453, 194, 487, 993, 34,
829, 317, 865, 296, 197, 895, 208, 613, 98, 487, 963, 81,
808, 182, 5, 869, 291, 549, 489, 49, 941, 473, 116, 705,
340, 209, 547, 156, 735, 573, 234, 259, 704, 711, 892,
509, 680, 280, 819, 385, 61[illegible], 666, 599, 389, 229, 862,
288, 971, 656, 18, 774, 22[illegible], 990, 786, 828, 605

Mean of sample k

$\bar{x}_1 = 555.2$

$\bar{x}_2 = 439.5$

$\bar{x}_3 = 625.3$

Sampling

Mean of sample k

985, 978, 435, 389, 7[illegible]26, 299, 538, 571, 828, 681,
302, 13, 518, 873, 256[illegible]9, 864, 314, 470, 547, 440,
699, 867, 860, 202, [illegible]55, 792, 64, 406, 906, 859, 584,
375, 996, 466, 401, 428, 714, 453, 194, 487, 993, 34,
829, 317, 865, 296, 197, 895, 208, 613, 98, 487, 963, 81,
808, 182, 5, 869, 291, 549, 489, 49, 941, 473, 116, 705,
340, 209, 547, 156, 735, 573, 234, 259, 704, 711, 892,
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288, 971, 656, 18, 774, 226, 990, 786, 828, 605

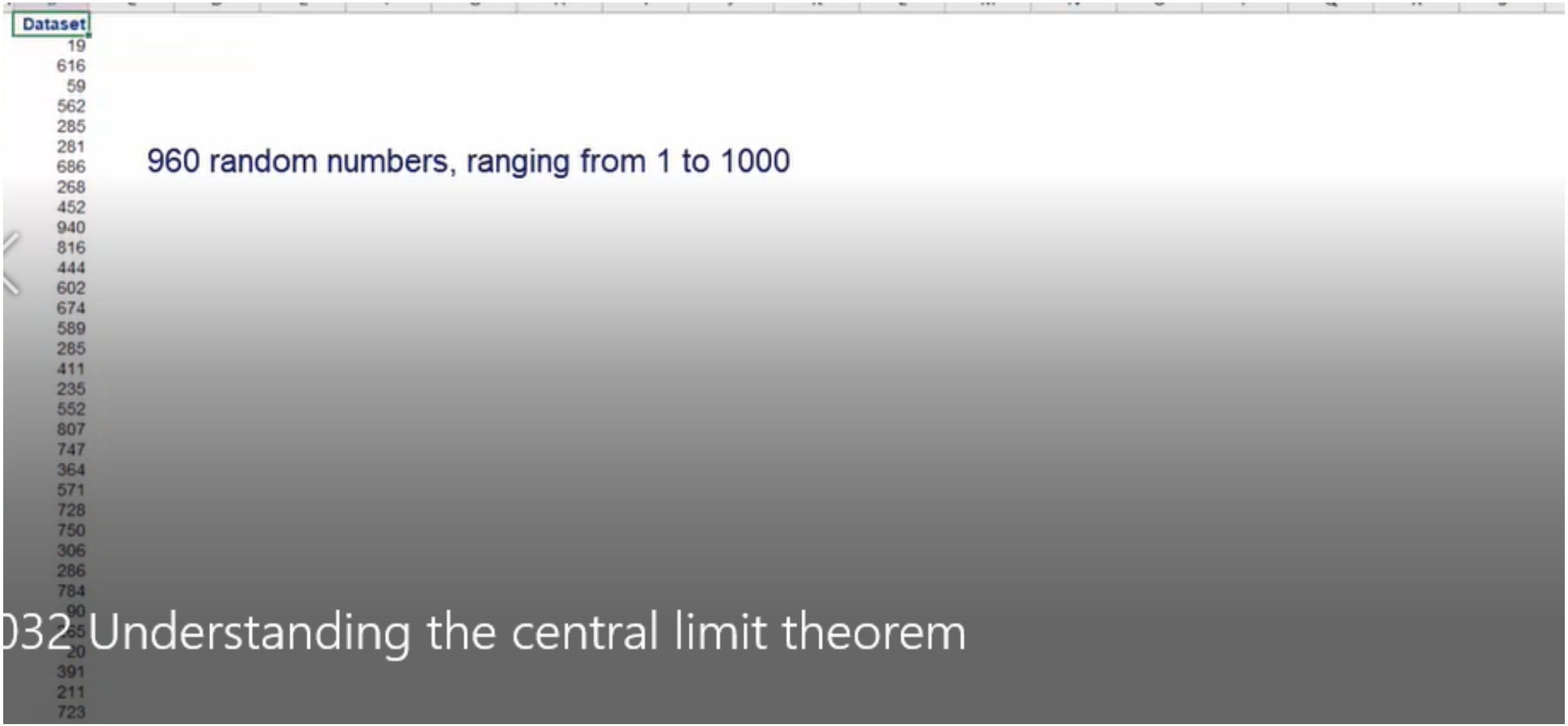
any distribution

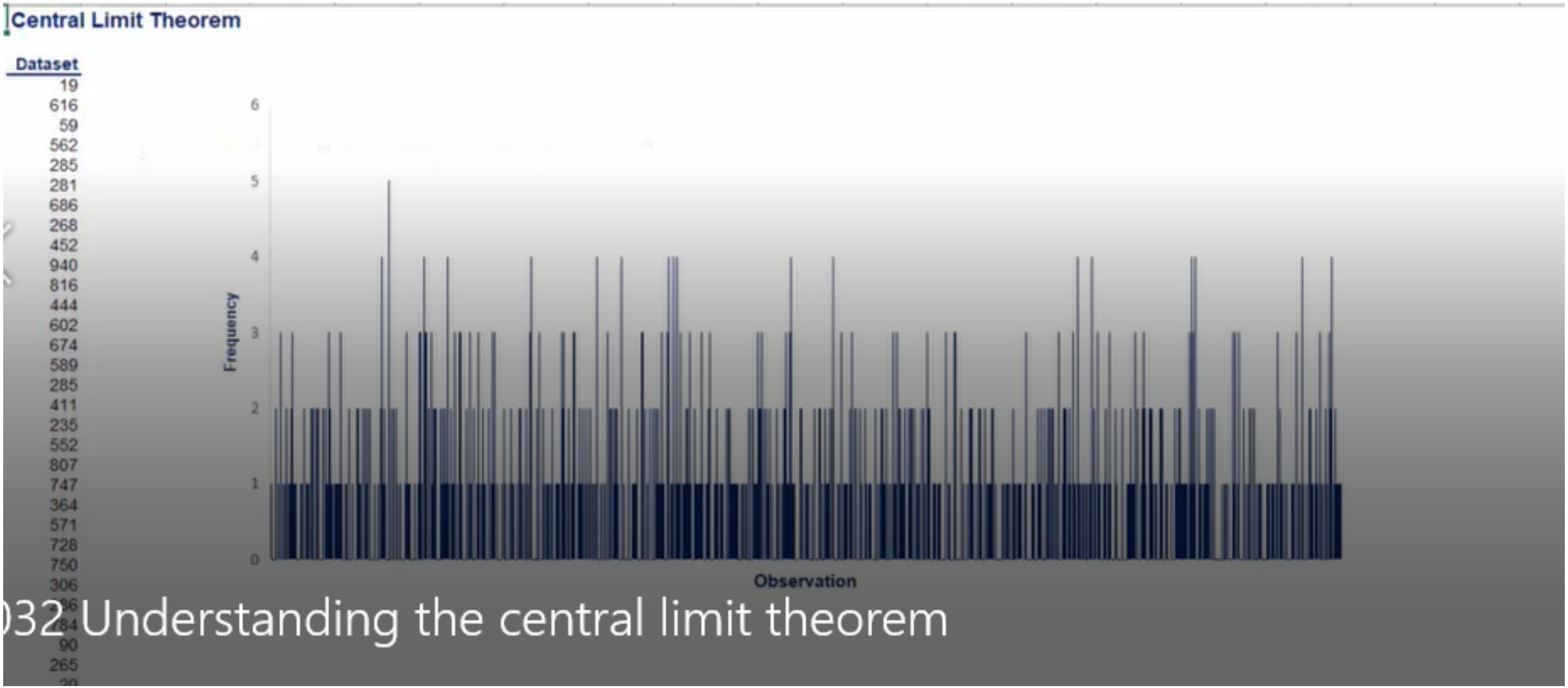
$\bar{x}_1 = 555.2$

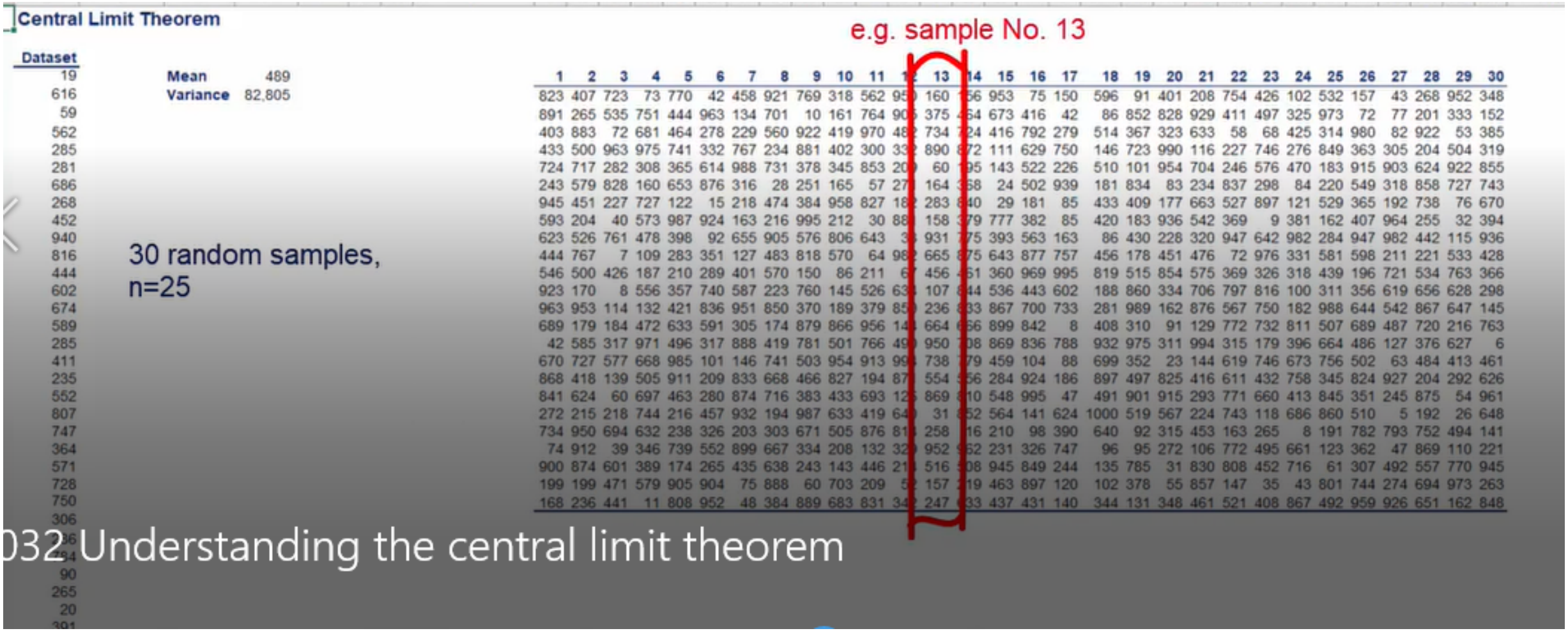
$\bar{x}_2 = 439.5$

$\bar{x}_3 = 625.3$

$\bar{x}_k = 567.5$


Dataset
19
616
59
562
285
281
686
268
452
940
816
444
602
674
589
285
411
235
552
807
747
364
571
728
750
306
286
784
391
211
723
960 random numbers, ranging from 1 to 1000
032 Understanding the central limit theorem


Central Limit Theorem
Dataset
19
616
59
562
285
281
686
268
452
940
816
444
602
674
589
285
411
235
552
807
747
364
571
728
750
306
90
265
Frequency
0
1
2
3
4
5
6
Observation
032 Understanding the central limit theorem


Central Limit Theorem
e.g. sample No. 13
Dataset
Mean 489
Variance 82,805
30 random samples,
n=25
Understanding the central limit theorem

Central Limit Theorem

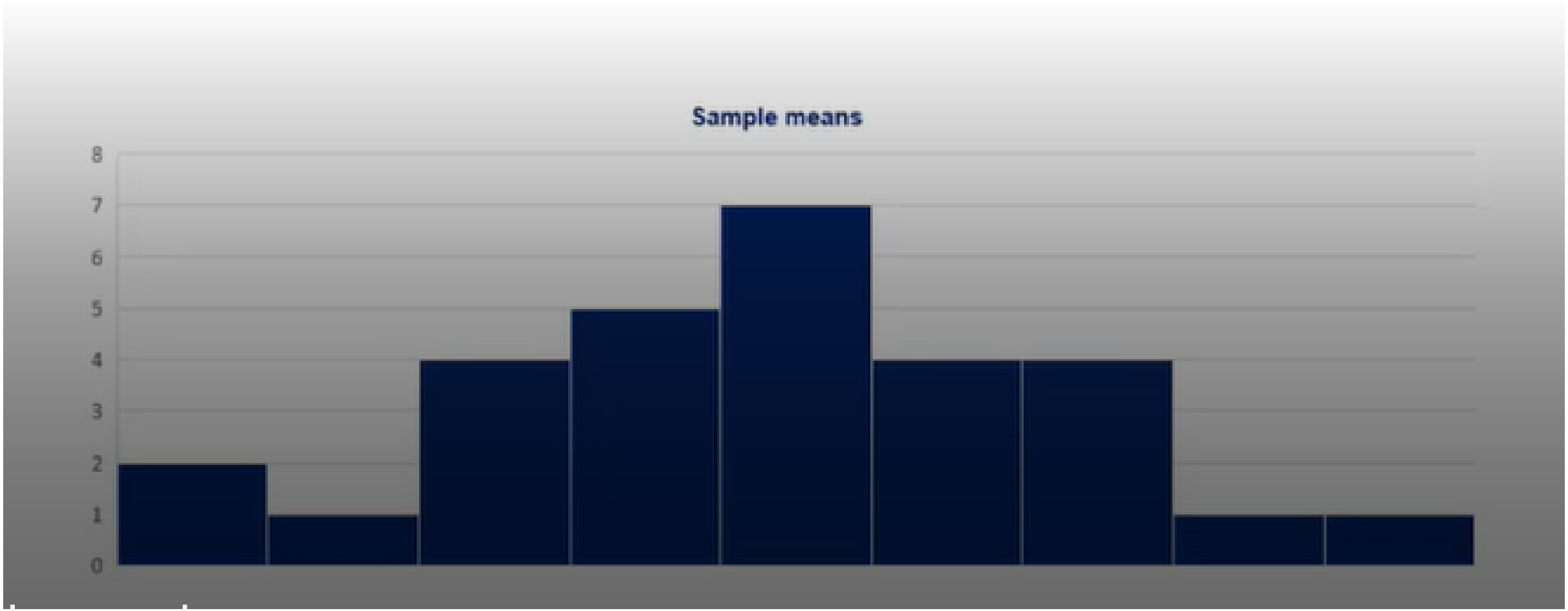
Dataset
19
616
59
562
285
281
686
268
452
940
816
444
602
674
589
285
411
235
552
807
747
364
571
728
750
306
90
265
20
391

Mean	489
Variance	82,805

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
823	407	723	73	770	42	458	921	769	318	562	950	160	156	953	75	150	596	91	401	208	754	426	102	532	157	43	268	952	348
891	265	535	751	444	963	134	701	10	161	764	905	375	464	673	416	42	86	852	828	929	411	497	325	973	72	77	201	333	152
403	883	72	681	464	278	229	560	922	419	970	482	734	724	416	792	279	514	367	323	633	58	68	425	314	980	82	922	53	385
433	500	963	975	741	332	767	234	881	402	300	332	890	872	111	629	750	146	723	990	116	227	746	276	849	363	305	204	504	319
724	717	282	308	365	614	988	731	378	345	853	209	60	195	143	522	226	510	101	954	704	246	576	470	183	915	903	624	922	855
243	579	828	160	653	876	316	28	251	165	57	271	164	368	24	502	939	181	834	83	234	837	298	84	220	549	318	858	727	743
945	451	227	727	122	15	218	474	384	958	827	182	283	840	29	181	85	433	409	177	663	527	897	121	529	365	192	738	76	670
593	204	40	573	987	924	163	216	995	212	30	881	158	379	777	382	85	420	183	936	542	369	9	381	162	407	964	255	32	394
623	526	761	478	398	92	655	905	576	806	643	38	931	775	393	563	163	86	430	228	320	947	642	982	284	947	982	442	115	936
444	767	7	109	283	351	127	483	818	570	64	982	665	875	643	877	757	456	178	451	476	72	976	331	581	598	211	221	533	428
546	500	426	187	210	289	401	570	150	86	211	67	456	461	360	969	995	819	515	854	575	369	326	318	439	196	721	534	763	366
923	170	8	556	357	740	587	223	760	145	526	634	107	844	536	443	602	188	860	334	706	797	816	100	311	356	619	656	628	298
963	953	114	132	421	836	951	850	370	189	379	850	236	833	867	700	733	281	989	162	876	567	750	182	988	644	542	867	647	145
689	179	184	472	633	591	305	174	879	866	956	144	664	666	899	842	8	408	310	91	129	772	732	811	507	689	487	720	216	763
42	585	317	971	496	317	888	419	781	501	766	490	950	708	869	836	788	932	975	311	994	315	179	396	664	486	127	376	627	6
670	727	577	668	985	101	146	741	503	954	913	994	738	779	459	104	88	699	352	23	144	619	746	673	756	502	63	484	413	461
868	418	139	505	911	209	833	668	466	827	194	871	554	556	284	924	186	897	497	825	416	611	432	758	345	824	927	204	292	626
841	624	60	697	463	280	874	716	383	433	693	126	869	810	548	995	47	491	901	915	293	771	660	413	845	351	245	875	54	961
272	215	218	744	216	457	932	194	987	633	419	640	31	852	564	141	624	1000	519	567	224	743	118	686	860	510	5	192	26	648
734	950	694	632	238	326	203	303	671	505	876	814	258	16	210	98	390	640	92	315	453	163	265	8	191	782	793	752	494	141
74	912	39	346	739	552	899	667	334	208	132	329	952	962	231	326	747	96	95	272	106	772	495	661	123	362	47	869	110	221
900	874	601	389	174	265	435	638	243	143	446	214	516	508	945	849	244	135	785	31	830	808	452	716	61	307	492	557	770	945
199	199	471	579	905	904	75	888	60	703	209	52	157	219	463	897	120	102	378	55	857	147	35	43	801	744	274	694	973	263
168	236	441	11	808	952	48	384	889	683	831	342	247	633	437	431	140	344	131	348	461	521	408	867	492	959	926	651	162	848

032 Understanding the central limit theorem

Rule of thumb: the sample should be bigger than 25 observations


Sample means
8
7
6
5
4
3
2
1
0